

A 60 GHz Millimeter-Wave FMCW Radar System for Real-Time Observations in Remote Outdoor Environments

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Abstract—Single-chip millimeter-wave radars provide opportunities for low-cost, high-resolution environmental monitoring applications (e.g., precipitation and snowpack characterization). However, the development of such applications is hindered by the lack of weatherproofing and remote operation capabilities of radar development hardware. In this paper, we present a low-cost, weatherproof 60 GHz frequency-modulated continuous wave radar system designed for environmental observations. The system relies on an existing Infineon BGT60TR13C development platform with WiFi connectivity, a radome, and a solar power source, facilitating autonomous operation, real-time configurability, and data acquisition. We assess the radome's impact on the radar's radiation pattern and demonstrate the system's reliability and suitability for environmental research through a year-long deployment in a mountainous watershed near Crested Butte, CO, USA.

Index Terms—Antenna radiation patterns, environmental engineering, millimeter wave propagation, FMCW radar, radome

I. INTRODUCTION

Recent developments of Frequency-Modulated Continuous-Wave (FMCW) radars in millimeter-wave (mmWave) bands have resulted in the widespread availability of radar systems for automotive and industrial applications [1]. These low-cost, single-chip solutions provide a high range resolution, low power consumption, and low hardware complexity, facilitating the introduction of the technology in novel domains of research [2], [3]. Environmental science is a particular field of research where mmWave radars can drive significant advances. Traditional sensing methods are often constrained by high equipment costs, complex deployment procedures, limited data accessibility, and inadequate capabilities for observing various environmental processes at high spatial and temporal resolutions [4]. To overcome these limitations, we envision a concept of low-cost, distributed mmWave radar sensors with wireless connectivity for real-time environmental monitoring in remote field sites. Previous environmental research has demonstrated the suitability of radar technology to study snowpack [5], [6], vegetation [7], snow-water equivalent [8],

soil water content [9], precipitation [10], and wind [11]. However, the use of mmWave radar technology in environmental research is often restricted to short-term studies due to power, weatherproofing, and data accessibility concerns. In this work, we present a robust, easily reproducible radar system designed for long-term outdoor deployments. The system is remotely controllable and configurable, and provides real-time access to in-phase and quadrature (I/Q) samples for environmental research. We particularly focus on the system's radome design and its influence on the radar's radiation patterns. We deployed multiple of these mmWave sensor stations at a mountainous watershed near Crested Butte, CO, USA in October 2023 and they have remained operational up to present date (May 2025). The presented solution relies on commercially available hardware and enables environmental scientists to acquire long-term datasets and dynamically adjust radar configurations in real time, allowing them to focus on data processing rather than hardware development.

This paper is organized as follows: Section II describes the hardware architecture, the radome design, and the software framework. Section III presents the system's overall performance, highlighting the radome's impact, and demonstrating environmental monitoring capabilities. Section IV contains the conclusions of our work.

II. METHODS

The requirements for a deployable mmWave radar system for environmental research include weatherproofing, robust software, configurable hardware, standalone power system, and wireless communication capabilities. Fig. 1 presents a block diagram of the proposed system designed to meet these needs. At its core, the system features a mmWave radar with an embedded WiFi module and a solar-based power system, both housed in weatherproof enclosures. Wireless communications infrastructure (satellite or cellular based), along with cloud servers for remote control and data storage, support the operation of the system.

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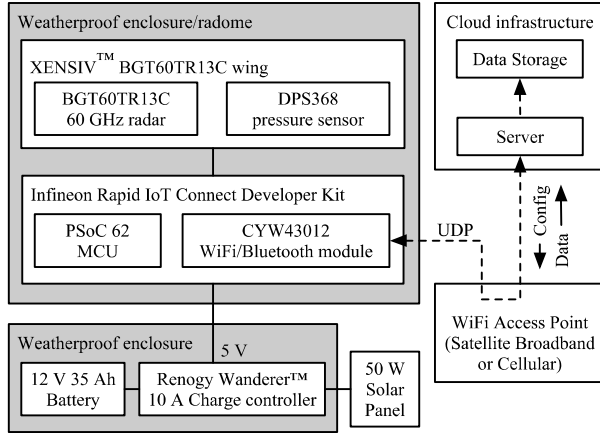


Fig. 1: Block diagram of the proposed mmWave radar system.

A. Hardware Architecture

The mmWave FMCW radar system must satisfy several criteria, including a low cost, low power consumption, minimal hardware complexity, compact size, straightforward programming and configuration, and wireless communication capabilities. We selected the 60 GHz FMCW radar prototyping kit from Infineon [12], which meets all of the aforementioned criteria. The platform contains a BGT60TR13C radar [13] with 4 on-package antennas (3 to receive (Rx) and 1 to transmit (Tx)), a PSoC 62 microcontroller (MCU) with an ARM Cortex-M4 and M0+, a CYW43012 WiFi and Bluetooth module, and a DPS368 barometric pressure sensor. The radar platform has an average power consumption of 1,200 mW during operation and less than 20 mW in sleep mode [14]. For power provisioning, we selected a 12 V, 35 Ah sealed lead acid battery (e.g., ML35-12) and a Renogy 50 W monocrystalline solar panel with a 10 A Wanderer™ pulse width modulation charge controller. This selection was driven by the need for system uptime during weeks of solar panel snow cover, operation at sub-zero °C temperatures (ruling out lithium-ion technology), and system cost. Assuming a total of 6 hours of radar operation per day (25% duty cycle), this solution ensures 25 days of operation in the absence of sunlight.

B. Radome Design

The battery and charge controller can be housed in a general purpose UV-resistant and waterproof enclosure, but the radome should also be lightweight and easily mountable, have a high albedo to minimize heating, and be transparent to mmWave radar and WiFi signals. White polypropylene (PP) was identified as a suitable enclosure material with good commercial availability. It has a relative permittivity of $\epsilon_r = 2.3$, and a low loss tangent of $\tan \delta = 10^{-4}$ at 60 GHz [15]. To minimize losses, the optimal enclosure thickness (t_{opt}) and the optimal antenna distance to the enclosure's wall (d_{opt}) are defined in (1) and (2) [16], [17].

$$t_{\text{opt}} = \frac{n\lambda_{\text{PP}}}{2} = n \cdot 1.65 \text{ mm}, \quad n \in \mathbb{N}, \quad (1)$$

$$d_{\text{opt}} = \frac{m\lambda_{\text{air}}}{2} = m \cdot 5 \text{ mm}, \quad m \in \mathbb{N}, \quad (2)$$

where c denotes the speed of light, n and m are integers, and the wavelength λ is given by (3):

$$\lambda = \frac{c}{f\sqrt{\epsilon_r}}. \quad (3)$$

Spherical radomes over a centered antenna exhibit a minimal impact on electromagnetic waves due to the consistently perpendicular incident angles to the radome, constant material thickness, and uniform distance to the radome over the entire field of view [16], [17]. However, given the rectangular shape of the radar module (63x26x22 mm), the aforementioned mechanical requirements, and the shape constraints of off-the-shelf enclosures, we selected the cylindrical PA0530500 jar with an outer diameter of 49.7 mm and its corresponding lid, manufactured by Parkway Plastics. Fig. 2a and 2b show the top and front views of the radar system assembly, and Fig. 2c displays the exploded view. From left to right, one can distinguish a desiccant packet, the radar platform, a 3D-printed polylactic acid (PLA) radar mount with 100% fill, and the enclosure with a power feed-through. The radar mount features an opening for the antennas, and aligns the radar chip at the center of the enclosure. The opening is designed as a revolving parabola ($y = 0.021 x^2$ [mm]) around the radar chip's 0° elevation and azimuth axis (boresight), preventing signal resonance between the cylinder's bases while limiting the amount of dielectric material (PLA) near the antennas. This revolved cut results in elevation and azimuth opening angles of $\pm 56.6^\circ$ and $\pm 60.3^\circ$, respectively (Fig. 3). The combination of the cylindrical shape of the radome and the placement of multiple off-center antennas on the radar chip, makes it impossible to fulfill the optimal thickness (t_{opt}) and distance (d_{opt}) requirements for all antennas under all angles of incidence. From the center of the enclosure, the ideal thickness is achieved at an elevation angle of $\pm 28^\circ$, while the optimal distances to the casing are achieved at elevation angles of $\pm [21, 39, 48, 54]^\circ$. To evaluate the impact of the enclosure, we measured the radiation patterns of all three Tx-Rx antenna combinations with and without radar enclosure. We captured the reflectivity from a corner reflector positioned 1 m away while rotating the radar along the azimuth (7.5° steps) and elevation (11.25° steps) axes. For these tests, the receiver gain was set to 33 dB and the bandwidth was maximized to 5.5 GHz, yielding a range resolution of 1.36 cm. To minimize noise, the results are averaged across all three Tx-Rx combinations.

C. Software

Our software solution contains embedded software, a server application, cloud storage, and post-processing scripts. The embedded software runs on the radar system's MCU, which controls the radar system and establishes WiFi connectivity to the server over UDP. Over this connection, the MCU receives the radar's configuration from the server (downlink), but it also uses the link to stream raw radar data to the server (uplink).

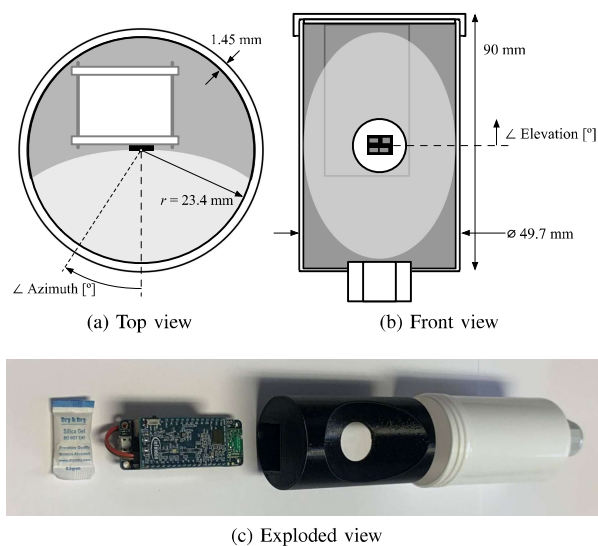
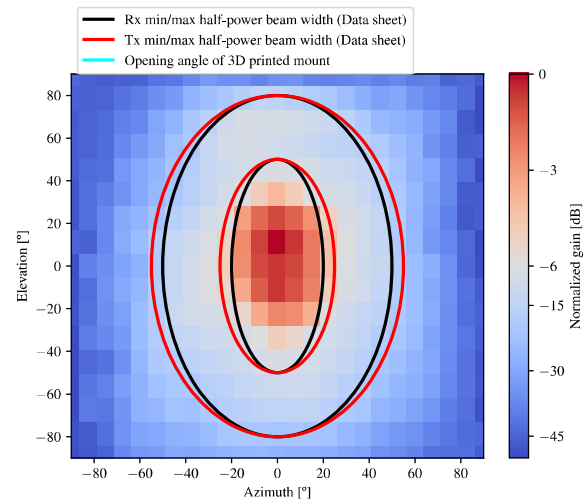


Fig. 2: Mechanical design of the radar assembly.

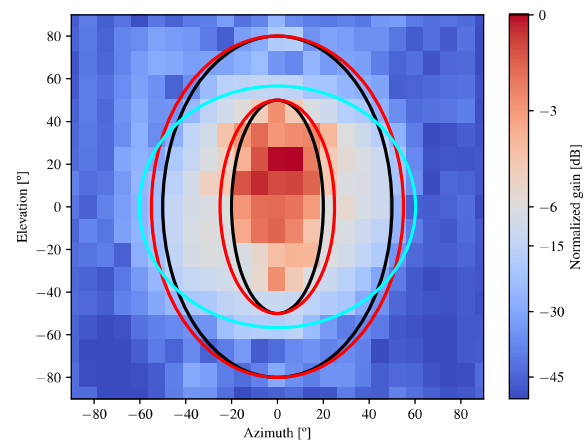
of I/Q samples). To establish the robustness and reliability needed for remote deployments several recovery mechanisms were implemented, including automatic reconnection attempts and a 24 h watchdog timer. The server application handles incoming data streams, but also remotely controls the radar's state (measure, sleep, or reset) and configuration. This enables dynamic remote management of the system's power consumption and performance, and measurement scheduling to avoid interference between radars. The radar configuration involves bandwidth, chirp duration, frame size, and signal power settings, affecting the resolution and maximum values of range and velocity measurements. The ideal configuration depends on the radar's application, orientation, and environment. Each measurement file contains all the information required for post-processing, including radar configuration data, time stamps, and I/Q samples for all radar frames and Rx antennas (with each radar frame consisting of multiple chirps). To enable long-term deployments, these files are automatically stored in an unlimited cloud data repository. Users can access this repository to download and post-process measurements. Processing scripts are application specific, but generally contain code for reflectivity, range, and velocity calculations, as well as frame corruption detection.

III. RESULTS AND DISCUSSION

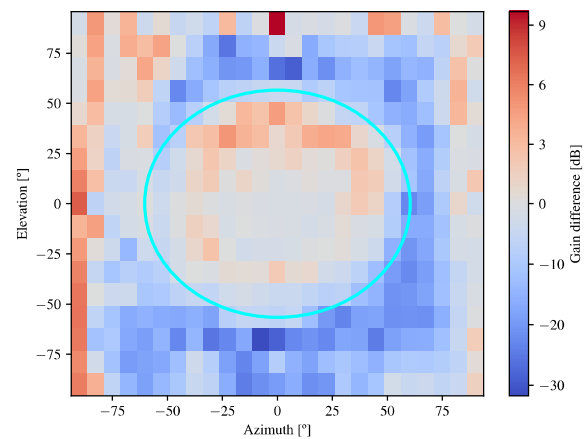
Fig. 3a and 3b show the radar's normalized radiation patterns without and with the enclosure, respectively. Since we assess the round-trip response from Tx-Rx pairs, the averaged single antenna half-power beam width is found at the -6 dB point. Using linear interpolation of the measurements from a bare radar module, the -6 dB points are identified at azimuth angles of $[-29^\circ, 29^\circ]$ and elevation angles of $[-39^\circ, 40^\circ]$. These results match the reported half-power beam widths in the radar's data sheet [13]. With the enclosure, the



(a) Gain pattern of the bare radar module (normalized)



(b) Gain pattern with enclosure (normalized)



(c) Gain difference after adding the enclosure

Fig. 3: Measured round-trip antenna gain patterns characterizing the impact of the enclosure (averaged over all Rx-Tx combinations).

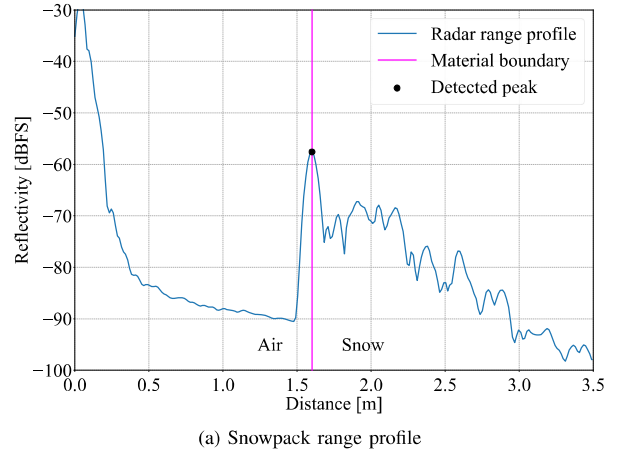


Fig. 4: Two radars deployed for real-time snowpack (a) and precipitation monitoring (b) near Crested Butte, CO, USA at 3,150 m elevation.

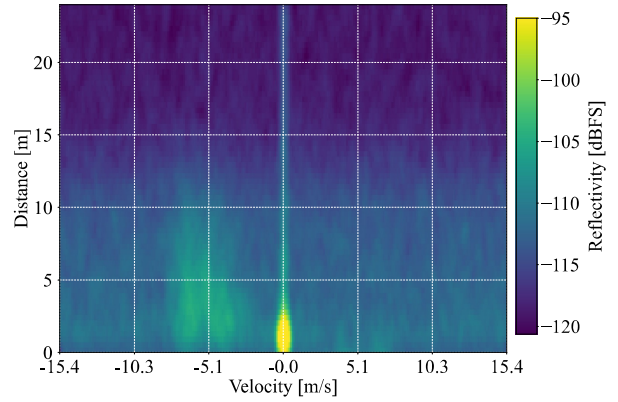
pattern remains within specifications with -6 dB points at $[-33^\circ, 35^\circ]$ azimuth angles and $[-39^\circ, 50^\circ]$ elevation angles. This demonstrates how the 3D-printed mount widens the beam in the azimuth domain, resulting in a slightly less eccentric beam shape. Outside the antenna mount's opening angles, round-trip signals are generally attenuated by at least 30 dB. Fig. 3c displays the absolute impact of the radome and radar mount. Despite the non-ideal wall thickness and distance to the cylindrical enclosure at boresight, absorption and reflection losses (0.54 dB) are negligible compared to the near-boresight measurement variability (± 0.8 dB). While the cylindrical design theoretically causes localized losses, these are overshadowed by noise and the dominant beam shaping effect of the 3D-printed mount; signals outside the mount's opening angle are strongly attenuated, while peripheral signals within the field of view get amplified, resulting in a slightly less eccentric beam shape, as observed in Fig. 3b.

In October 2023, we deployed 6 autonomous radar systems in a mountainous watershed near Crested Butte, CO, USA (Fig. 4), equipped with Starlink and cellular WiFi access points. The systems were exposed to a large range of temperatures (-22.9°C to 25.1°C), low atmospheric pressure (average 726.7 hPa), low humidity, high winds, and elevated cosmic rays and electrostatic buildup. Downward-facing radars are used for snowpack observations, while horizontally and upwards facing units measure precipitation. We installed solar panels at suboptimal high inclination angles to reduce snow accumulation, which avoided outages throughout the 16-month deployment. During this period, we acquired more than 600,000 real-time observations under various configurations, each resulting in a 3 MB to 35 MB file. This represents 93% of the expected observations, with no radar downtime exceeding 24 consecutive hours. Reducing the watchdog timer from 24 to 2 hours could have boosted the success rate to 99%.

Figure 5a illustrates the ability of a downward-facing radar to accurately measure the distance to the snowpack (here



(a) Snowpack range profile



(b) Range-velocity plot for rain

Fig. 5: Demonstration of environmental radar observations.

160 cm). Fig. 5b displays a range-velocity plot from an upward-facing radar during a precipitation event, measuring hydrometeor velocities between 3.1 m/s and 7.4 m/s.

IV. CONCLUSION

We presented a weatherproof FMCW mmWave radar system with wireless connectivity for real-time observations in remote environments. The system features a BGT60TR13C based development platform, centered in a cylindrical PP enclosure. We measured the impact of this enclosure on the radar's radiation pattern, demonstrating that the radome has a negligible influence, while the 3D printed radar mount attenuates the signals beyond its opening angles, and creates a less eccentric radar beam shape. Six stations were deployed at a mountainous field site and remained operational for more than a year. Snowpack and precipitation measurements demonstrated the system's suitability for environmental research. Our future research will focus on the development of dedicated post-processing algorithms for various environmental processes.

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